

// Day 41 of 60

SYSTEM DESIGN

Designing a URL Shortener

→ Swipe through all 6 slides



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The Core Problem

From long path to 7 characters

Long URL:

`https://example.com/very/long/path`

Short URL:

`https://sh.rt/XyZ123`

// Goal: shrink any URL to 7 chars
// while staying unique forever



7 characters, 3.5 trillion possibilities



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Base62 Encoding

Where the 7 characters come from

Charset: A-Z, a-z, 0-9
→ 62 possible characters

7 characters → 62^7
= 3,521,614,606,208
≈ 3.5 trillion unique URLs

// Enough for every human
// to shorten 400+ URLs each



Avoid 0/O and 1/l for readability



Database Design

Keep the schema dead simple

Table: short_urls (NoSQL)

```
short_url    PRIMARY KEY
original_url
created_at
expires_at
user_id      (optional)
```

```
// Cassandra or DynamoDB
// Single-key lookup = fast
```



No joins. No relations. Just KV lookups.



Generating the Unique ID

Avoiding collisions across servers

Problem: 2 servers → same short URL?

Solution: Twitter Snowflake

timestamp	machine	seq
-----------	---------	-----

- unique 64-bit number
- encode to Base62
- done.



Centralized ID generation prevents collisions



Read vs Write Ratio

URL shorteners are read-heavy

```
Writes (create short URL):    1
```

```
Reads  (redirect to long): 100
```

```
Ratio: 100:1
```

```
// Optimize for reads
```

```
// Cache aggressively
```

```
// Don't over-engineer writes
```



Caching is your best friend here



Redis Caching Layer

Sub-10ms redirects at scale

```
GET /XyZ123
```

↓

```
Check Redis → HIT? return 301
```

↓ MISS

```
Check DB → return + cache it
```

↓

```
Response < 10ms
```

```
// TTL: 24 hours
```

```
// LRU eviction policy
```



Cache hit ratio of 95%+ is typical



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done = true

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